Unit 11, Lesson 2: Constructing the Incenter of a Triangle

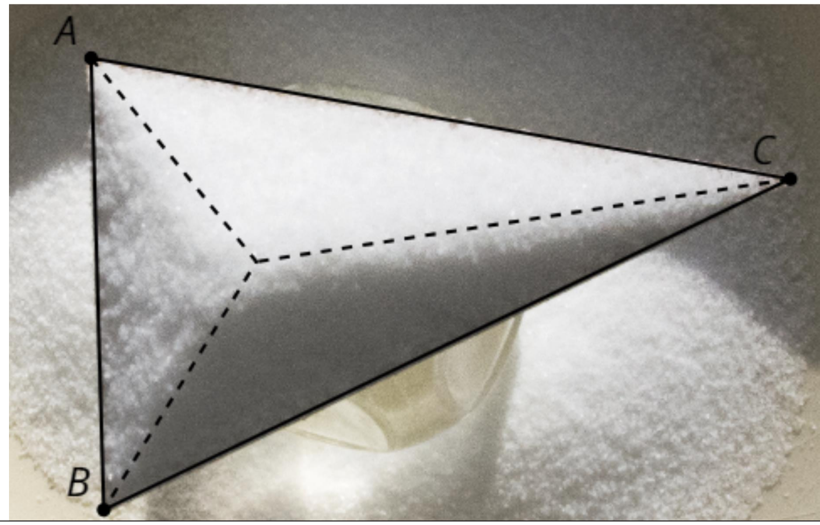
Benchmark

MA.912.GR.5.3 Construct the inscribed and circumscribed circles of a triangle.

Learning Targets

- ☐ I can construct an inscribed circle in a triangle.
- ☐ I can locate the incenter of a triangle.

11.2.1 Warm-Up: Salt Pile



Salt is poured onto a triangle until it piles up. As the salt piles up, the pile reaches a maximum height. As the salt reaches a maximum height, new grains of salt fall off whichever side of the triangle is closest.

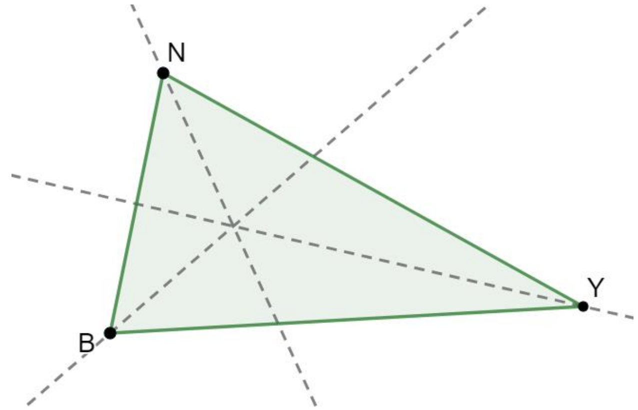
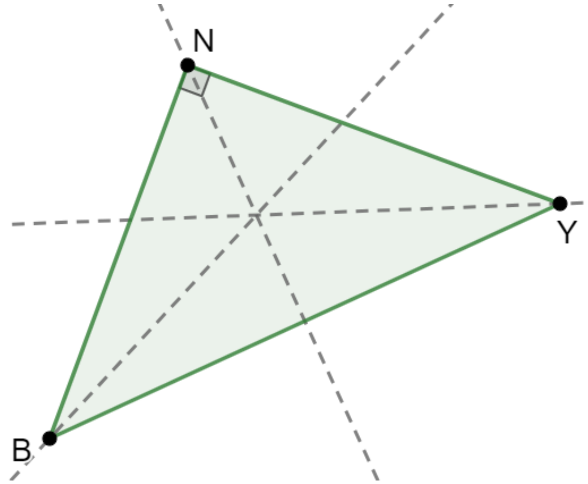
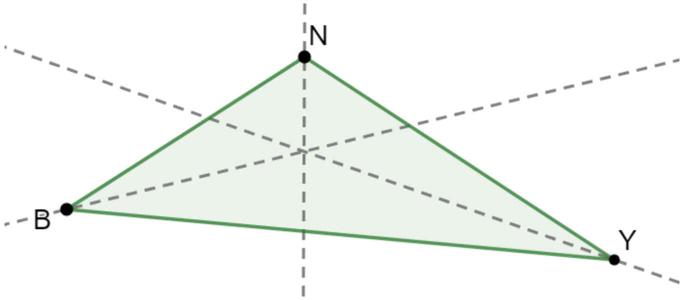
1. What do you wonder about the salt pile?

2. What do you notice about the salt pile?

3. What do you notice about the point where the salt reaches a maximum?

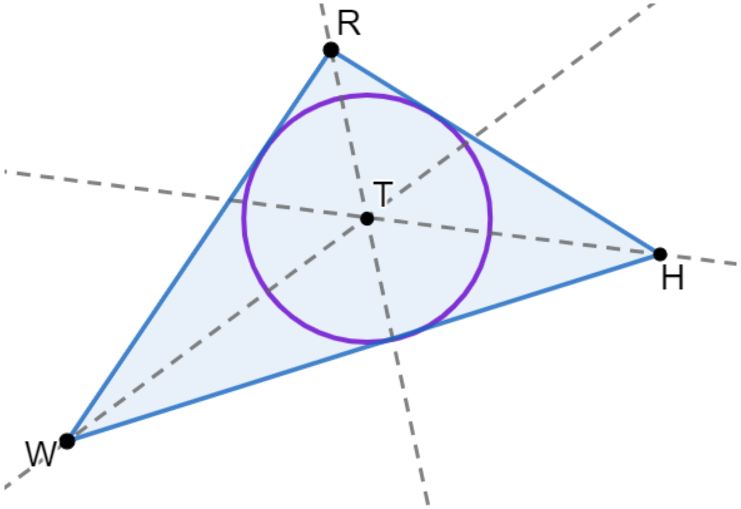
11.2.2 Exploration: Constructing an Inscribed Circle

1. ΔNYB is shown as three different types of triangles.
   a. Label each point of concurrency as point R.

Acute	
Right	
Obtuse	

b. What do you notice about the three different triangles and point R, the point of concurrency?

2. ΔRHW is shown where each angle is bisected. The three angle bisectors meet at point T.

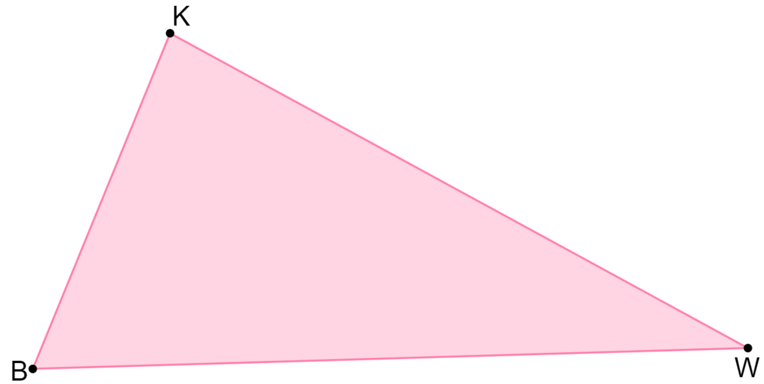


How is circle T related to ΔRHW?

3. Use a compass to draw the inscribed circles for each type of ΔNYB in problem 1.

The point of concurrency of the angle bisectors of a triangle is the ***incenter***.

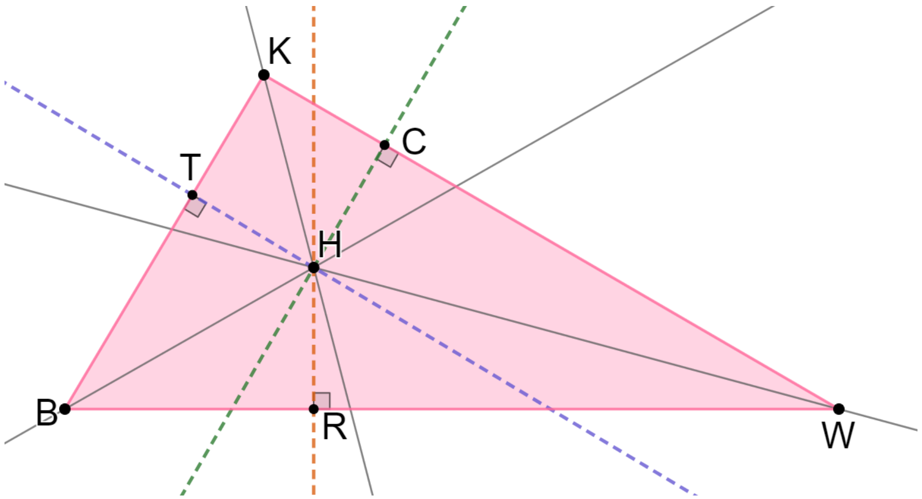
4. Consider ΔKWB.



a. Copy ΔKWB onto patty paper and find the incenter of the triangle by constructing the angle bisectors. Label the incenter of ΔKWB as point H.

b. Draw the inscribed circle for ΔKWB using point H, the incenter, as the center of the circle.

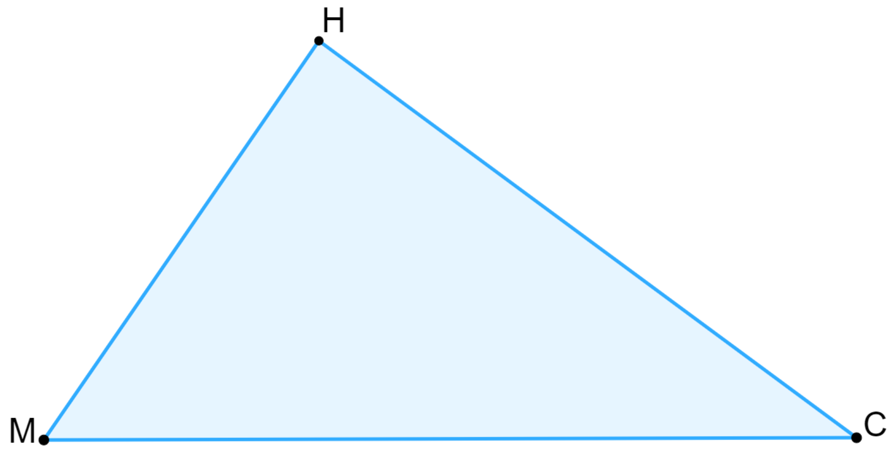
c. Noah is concerned that when he draws the inscribed circle, he does not have the exact points on the triangle where the circle intersects the triangle. He drew three perpendicular lines from the center to each side.



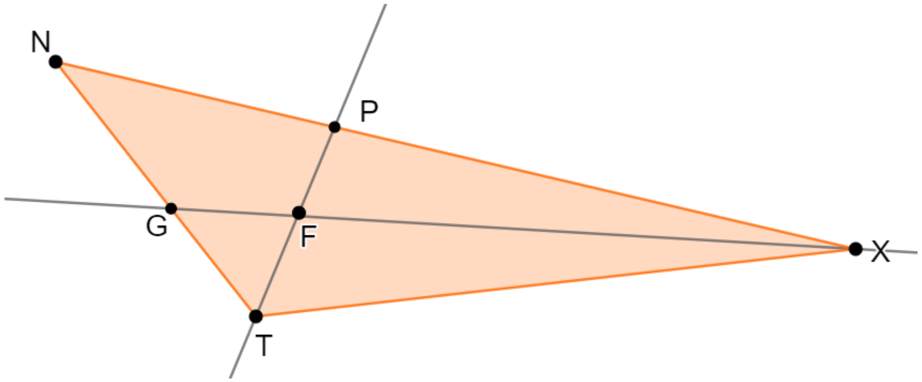
When Circle H is drawn on Noah's version of ΔKWB, what relationship do $\overline{HR}$, $\overline{HT}$, and $\overline{HC}$ have to each other and to Circle H?

d. Draw Circle H that passes through points T, C, and R.

5. Copy ΔMHC onto another piece of paper. Then, use a compass and a straight edge to inscribe a circle in ΔMHC.



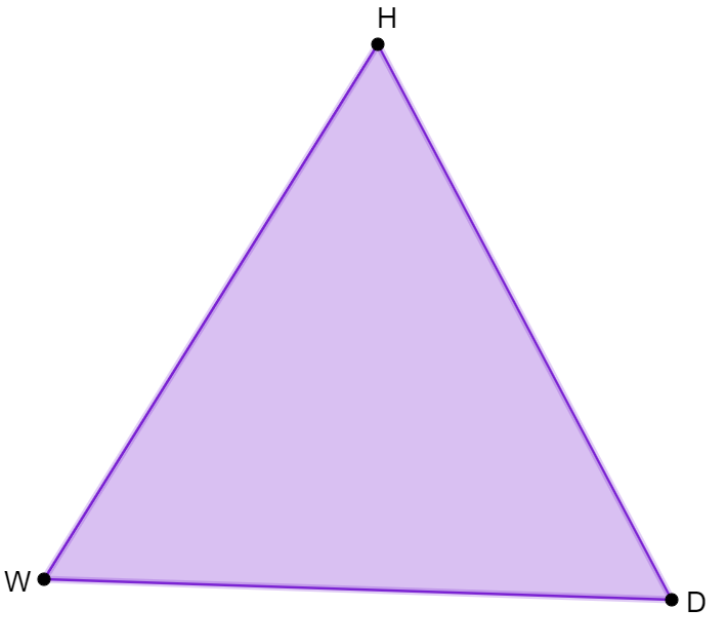
6. ΔNXT is shown where $\overline{PT}$ bisects $\angle NTX$ and $\overline{GX}$ bisects $\angle PXT$.



a. Will the inscribed circle of ΔNXT pass through points P and G?

b. Justify why $\overline{PF}$ and $\overline{FG}$ are not radii of the inscribed Circle F.

7. Copy ΔHDW onto a piece of patty paper. Then, construct the incenter and circumcenter of ΔHDW.



11.2.3 Wrap-Up: Reflection

1. What was the ***muddiest*** (most difficult or confusing) point in today's lesson on constructing a circle inscribed in a triangle?